Metrics detected in one RAG run

12 questions on gpt-5-mini; the app is the course RAG chatbot, traced by Langfuse. 16 of 19 Observability and 14 of 27 Evaluation metrics have a value.

Cells show what each tool reported in this run. Blank = the tool reported nothing for that metric. Judge: gpt-5-mini, unvalidated against human labels. Drawn by plot_matrix.py from the tools' output.

	Langfuse	Phoenix evals	DeepEval
Observability · Tool use			
Tool execution errors	0 of 10 tool calls		
Tool call count	10 in 12 turns		
Tool execution latency	median 18 ms		
Observability · Plan and trajectory			
Step count			
Observability · Token usage			
Input tokens	19,684		
Output tokens	2,598		
Total tokens	27,914		
Cached tokens	0		
Reasoning tokens	5,632		
Observability · Cost			
Estimated cost	$0.0214		
Cost breakdown by token type	in $0.0049 · out $0.0052 · reasoning $0.0113		
Observability · Latency			
Trace / run latency	median 8.8 s per turn		
LLM call latency	median 3.0 s		
Span duration	on every observation		
Time to first token			
Time per output token / generation time			
Observability · Errors and volume			
Error count / rate	0 error observations		
Trace / span count	12 traces, 44 observations		
LLM call count	22		
Evaluation · Tool use			
Tool selection		8/10 calls correct	
Tool argument validity		10/10 calls valid	mean 1.00 over 8 (ArgumentCorrectness)
Tool call correctness			mean 0.88 over 8 (ToolCorrectness)
Tool call efficiency			
Tool response handling		8/8 turns correct	
Evaluation · Task and goal			
Task / goal completion		10/12 turns complete	mean 0.71 over 12 (TaskCompletion)
Evaluation · Plan and trajectory			
Plan quality			
Plan adherence			
Trajectory accuracy			
Step / action efficiency			
Action progress toward goal			
Flow and reasoning coherence			
Evaluation · Answer quality			
Answer correctness			
Answer relevance			mean 0.92 over 12 (AnswerRelevancy)
Helpfulness			
Conciseness		1/12 turns concise	
Tone appropriateness			
Evaluation · Grounding			
Faithfulness to context		7/8 turns faithful	mean 1.00 over 8 (Faithfulness)
Evaluation · Safety and policy			
Toxicity		0/12 turns toxic	
Bias and fairness			mean 1.00 over 12 (Bias)
Sensitive data leakage		0/12 turns flagged	mean 0.99 over 12 (PIILeakage)
Injection detection			
Refusal detection		1/11 turns refused	
Policy and moderation violations			
Evaluation · Conversation			
Knowledge retention			mean 1.00 over 1 (KnowledgeRetention)
User satisfaction or frustration		1/12 turns with friction	
Conversation coherence			